Allomelanin-based biomimetic nanotherapeutics for orthotopic glioblastoma targeted photothermal immunotherapy.

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Immune checkpoint blockade (ICB) therapy has shown great potential in the treatment of malignant tumors, but its therapeutic effect on glioblastoma (GBM) is unsatisfactory because of the low immunogenicity and T cell infiltration, as well as the presence of blood-brain barrier (BBB) that blocks most of ICB agents to the GBM tissues. Herein, we developed a biomimetic nanoplatform of AMNP@CLP@CCM for GBM-targeted photothermal therapy (PTT) and ICB synergistic therapy by loading immune checkpoint inhibitor CLP002 into the allomelanin nanoparticles (AMNPs) and followed by coating cancer cell membranes (CCM). The resulting AMNP@CLP@CCM can successfully cross the BBB and deliver CLP002 to GBM tissues due to the homing effect of CCM. As a natural photothermal conversion agent, AMNPs are used for tumor PTT. The increased local temperature by PTT not only enhances BBB penetration but also upregulates the PD-L1 level on GBM cells. Importantly, PTT can effectively stimulate immunogenic cell death to induce tumor-associated antigen exposure and promote T lymphocyte infiltration, which can further amplify the antitumor immune responses of GBM cells to CLP002-mediated ICB therapy, resulting in significant growth inhibition of the orthotopic GBM. Therefore, AMNP@CLP@CCM has great potential for the treatment of orthotopic GBM by PTT and ICB synergistic therapy. STATEMENT OF SIGNIFICANCE: The effect of ICB therapy on GBM is limited by the low immunogenicity and insufficient T-cell infiltration. Here we developed a biomimetic nanoplatform of AMNP@CLP@CCM for GBM-targeted PTT and ICB synergistic therapy. In this nanoplatform, AMNPs are used as both photothermal conversion agents for PTT and nanocarriers for CLP002 delivery. PTT not only enhances BBB penetration but also upregulates the PD-L1 level on GBM cells by increasing local temperature. Additionally, PTT also induces tumor-associated antigen exposure and promotes T lymphocyte infiltration to amplify the antitumor immune responses of GBM cells to CLP002-mediated ICB therapy, resulting in significant growth inhibition of the orthotopic GBM. Thus, this nanoplatform holds great potential for orthotopic GBM treatment.

T cell receptor dynamic and transcriptional determinants of T cell expansion in glioma-infiltrating T cells.

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Glioblastoma (GBM) is characterized by low numbers of glioma-infiltrating lymphocytes (GIL) with a dysfunctional phenotype. Whether this dysfunctional phenotype is fixed or can be reversed upon ex vivo culturing is poorly understood. The aim of this study was to assess T cell receptor (TCR)-dynamics and -specificities as well as determinants of in vitro GIL expansion by sequencing-based technologies and functional assays to explore the use of GIL for cell therapy. By means of flow cytometry, T cell functionality in GIL cultures was assessed from 9 GBM patients. TCR beta sequencing (TCRB-seq) was used for TCR repertoire profiling before and after in vitro expansion. Microarrays or RNA sequencing (RNA-seq) were performed from 6 micro-dissected GBM tissues and healthy brain RNA to assess the individual expression of GBM-associated antigens (GAA). GIL reactivity against in silico predicted tumor-associated antigens (TAA) and patient-individual GAA was assessed by ELISpot assay. Combined ex vivo single cell (sc)TCR-/RNA-seq and post-expansion TCRB-seq were used to evaluate transcriptional signatures that determine GIL expansion. Human GIL regains cellular fitness upon in vitro expansion. Profound TCR dynamics were observed during in vitro expansion and only in one of six GIL cultures, reactivity against GAA was observed. Paired ex vivo scTCR/RNA-seq and TCRB-seq revealed predictive

transcriptional signatures that determine GIL expansion. Profound TCR repertoire dynamics occur during GIL expansion. Ex vivo transcriptional T cell states determine expansion capacity in gliomas. Our observation has important implications for the use of GIL for cell therapy including genetic manipulation to maintain both antigen specificity and expansion capacity.

Irritable Bowel Syndrome Associated with Blastocystis hominis or Without Relationship to It? A Case-Control Study and Minireview.

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Blastocystis hominis (B. hominis) is a protozoan parasite that has a worldwide distribution. Some studies have suggested a link between B. hominis and the development of irritable bowel syndrome (IBS). The objective of this study was to determine the prevalence of B. hominis in patients with IBS compared to healthy individuals. A total of 65 stool samples from patients with IBS and 65 samples from healthy individuals in northern Iran were examined. The samples were tested using various methods including direct smear, formalin ether sedimentation and culture to detect the presence of B. hominis. Additionally, polymerase chain reaction (PCR) was performed on all culture-positive isolates to confirm the results and identify the genotype. B. hominis was detected in 15.38% of IBS patients and 9.2% of the healthy group. The culture in RPMI1640 was found to be better than the formalin ether and direct smear methods. Positive samples were confirmed using the molecular method. No significant difference was observed in the order of B. hominis infection between the two groups. The results of our study indicate that no significant difference was observed in the order of B. hominis infection between IBS patients and healthy groups. Therefore, further study is necessary to determine the potential pathogenic effects of this parasite and its role in causing IBS.

Post-infectious irritable bowel syndrome following a diagnosis of traveller's diarrhoea: a comprehensive characterization of clinical and laboratory parameters.

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Prolonged or recurrent gastrointestinal symptoms may persist after acute traveller's diarrhoea (TD), even after adequate treatment of the primary cause. This study aims to describe the epidemiological, clinical and microbiological characteristics of patients with post-infectious irritable bowel syndrome (PI-IBS) after returning from tropical or subtropical areas. We conducted a retrospective study of patients presenting between 2009 and 2018 at the International Health referral centre in Barcelona with persistent gastrointestinal symptoms following a diagnosis of TD. PI-IBS was defined as the presence of persistent or recurrent gastrointestinal manifestations for at least 6 months after the diagnosis of TD, a negative stool culture for bacterial pathogens and a negative ova and parasite exam after targeted treatment. Epidemiological, clinical and microbiological variables were collected. We identified 669 travellers with a diagnosis of TD. Sixty-eight (10.2%) of these travellers, mean age 33 years and 36 (52.9%) women, developed PI-IBS. The most frequently visited geographical areas were Latin America (29.4%) and the Middle East (17.6%), with a median trip duration of 30 days (IQR 14-96). A microbiological diagnosis of TD was made in 32 of these 68 (47%) patients, 24 (75%) of whom had a parasitic infection, Giardia duodenalis being the most commonly detected parasite ($n = 20$, 83.3%). The symptoms persisted for a mean of 15 months after diagnosis and treatment of TD. The multivariate analysis revealed that parasitic infections were independent risk factors for PI-IBS (OR 3.0, 95%CI 1.2-7.8). Pre-travel counselling reduced the risk of PI-IBS (OR 0.4, 95%CI 0.2-0.9). In our cohort, almost 10% of patients with travellers' diarrhoea developed persistent symptoms compatible with PI-IBS. Parasitic infections, mainly giardiasis, seem to be associated with PI-IBS.

Gastrointestinal involvement in post-acute Coronavirus disease (COVID)-19 syndrome.

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Ten percentage of patients with coronavirus disease (COVID)-19 report having gastrointestinal (GI) symptoms as severe acute respiratory syndrome coronavirus-2 (SARS-CoV2) not only infects the pulmonary but also the GI tract. GI infections including that due to viral infection is known to cause postinfection disorders of gut-brain interaction (DGBI); hence, we wish to review the long-term GI consequences following COVID-19, particularly post-COVID-19 DGBI. At least 12 cohort studies, four of which also included controls documented the occurrence of post-COVID-19 DGBI, particularly IBS following COVID-19. The risk factors for post-COVID-19 DGBI included female gender, symptomatic COVID-19, particularly GI symptoms, the severity of COVID-19, the occurrence of anosmia and ageusia, use of antibiotics and hospitalization during the acute illness, persistent GI symptoms beyond 1 month after recovery, presence of mental health factors, The putative mechanisms for post-COVID-19 DGBI include altered gut motility, visceral hypersensitivity, gut microbiota dysbiosis, GI inflammation, and immune activation, changes in intestinal permeability, and alterations in the enteroendocrine system and serotonin metabolism. Long-term sequelae of SARS-CoV2 infection may persist even after recovery from COVID-19. Patients with COVID-19 are more likely to develop post-COVID-19 IBS than healthy controls. Post-COVID-19 IBS may pose a substantial healthcare burden to society.

Campylobacter infection and the link with Irritable Bowel Syndrome: on the pathway towards a causal association.

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proving causality between an exposure and outcome can be difficult in humans. Here, we utilize the Bradford Hill (BH) criteria to summarize the causal relationship between Campylobacter infection and the development of Irritable Bowel Syndrome (IBS). we utilized the BH criteria to assess the strength, consistency, specificity, temporality, biological gradient, plausibility, coherence, experiment, and analogy of the current evidence linking Campylobacter to IBS. Through a consensus amongst all authors, the confidence of each criterion was graded as high, moderate, low, or very low. a total of four criteria (strength, temporality, plausibility, and analogy) were graded as high; four criteria (consistency, biological gradient, coherence, and experiment) were graded as moderate; and one criterion (specificity) was graded as low. Large-scale epidemiological studies report a risk ratio of 2.7-5.6 for developing IBS after campylobacter. In rodent models, Campylobacter jejuni 81-176 can cause loose stool months after the infection is cleared and share common pathophysiology as IBS patients such as elevated intestinal TLR-4 and IL-8, antibodies to CdtB and vinculin, increased intraepithelial lymphocytes, and small intestinal bacterial overgrowth. Campylobacter infection appear to cause IBS in a subset of patients. This may hold implication in risk factor identification, public health policy, and possibly treatment.

The possible role of bacteria, viruses, and parasites in initiation and exacerbation of irritable bowel syndrome.

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Irritable bowel syndrome (IBS) is a prolonged and disabling functional gastrointestinal disorder with the incidence rate of 18% in the world. IBS could seriously affect lifetime of patients and cause high economic burden on the community. The pathophysiology of the IBS is hardly understood, whereas several possible mechanisms, such as visceral hypersensitivity, irregular gut motility, abnormal brain-gut relations, and the role of infectious agents, are implicated in initiation and development of this syndrome. Different studies demonstrated an alteration in B-lymphocytes, mast cells (MC), T-lymphocytes, and cytokine concentrations in intestinal mucosa or systemic circulation that are likely to contribute to the formation of the IBS. Therefore, IBS could be developed in those with genetic predisposition. Infections' role in initiation and exacerbation of IBS has been investigated by quite several clinical studies; moreover, the possible role of some pathogens in development and exacerbation of this disease has been described. It appears that the main obligatory pathogens correspond with the IBS disease, Clostridium difficile, Escherichia coli, Mycobacterium avium subspecies paratuberculosis, Campylobacter concisus, Campylobacter jejuni, Chlamydia trachomatis, Helicobacter pylori, Pseudomonas aeruginosa, Salmonella spp, Shigella spp, and viruses, particularly noroviruses. A number of pathogenic parasites (Blastocystis, Dientamoeba fragilis, and Giardia lamblia) may also be involved in the progression and exacerbation of the disease. Based on the current knowledge, the current study concludes that the most common bacterial, viral, and parasitic pathogens may be involved in the development and progression of IBS.

Giardia lamblia infection increases risk of chronic gastrointestinal disorders.

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Giardia lamblia is a common parasitic cause of infectious gastroenteritis in the United States and the world and may be linked to an increased risk of chronic gastrointestinal (GI) disorders. We sought to assess the risk of several chronic GI disorders following Giardia infection among active duty US military personnel. This study was designed as a retrospective cohort study in which active duty military personnel with documented G. lamblia infection were assessed for the subsequent risk of developing a chronic GI disorder including irritable bowel syndrome (IBS), dyspepsia and gastroesophageal reflux disease (GERD). Post-giardia chronic GI disorder risk was compared to risk in uninfected personnel matched on several demographic characteristics and medical encounter information. Data were obtained from the Defense Medical Surveillance System and exposures (1998-2009) with outcomes identified based on documented medical encounters with specific medical billing codes. Modified Poisson regression was used to evaluate the relationship between G. lamblia infection and chronic GI disorders. A total of 80 Giardia cases were identified for an estimated incidence of 0.55 cases per 100,000 person-years. Cases were matched to 294 unexposed subjects. After adjusting for important covariates, there was an increased risk of IBS (relative risk: 2.1, $p = 0.03$) associated with antecedent Giardia infection. These data add to a growing body of literature and demonstrate an increased risk of IBS after infection with G. lamblia.

Is irritable bowel syndrome an infectious disease?

PMID: 26819502 | Indexed in PubMed

Irritable bowel syndrome (IBS) is the most common of all gastroenterological diseases. While many mechanisms have been postulated to explain its etiology, no single mechanism entirely explains the heterogeneity of symptoms seen with the various phenotypes of the disease. Recent data from both basic and clinical sciences suggest that underlying infectious disease may provide a unifying hypothesis that better explains the overall symptomatology. The presence of small intestinal bowel overgrowth (SIBO) has been documented in patients with IBS and reductions in SIBO as determined by breath testing correlate with IBS symptom improvement in clinical trials. The incidence of new onset IBS symptoms following acute infectious gastroenteritis also suggests an infectious cause. Alterations in microbiota-host interactions may compromise epithelial barrier integrity, immune function, and the development and function of both central and enteric nervous systems explaining alterations in the brain-gut axis. Clinical evidence from treatment trials with both probiotics and antibiotics also support this etiology. Probiotics appear to restore the imbalance in the microflora and improve IBS-specific quality of life. Antibiotic trials with both neomycin and rifaximin show improvement in global IBS symptoms that correlates with breath test normalization in diarrhea-predominant patients. The treatment response to two weeks of rifaximin is sustained for up to ten weeks and comparable results are seen in symptom reduction with retreatment in patients who develop recurrent symptoms.

Role of gut pathogens in development of irritable bowel syndrome.

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Acute infectious gastroenteritis is one of the most commonly identifiable risk factors for the development of irritable bowel syndrome (IBS). A number of bacterial, viral and parasitic pathogens have been found to be associated with the development of IBS and other functional gastrointestinal (GI) disorders. Epidemiological studies have identified demographic and acute enteritis-related risk factors for the development of post-infectious-IBS (PI-IBS). Immune dysregulation, alterations in barrier function, serotonergic and mast cell activation have been identified as potential pathophysiological mechanisms. Additionally, variations in host genes involved in barrier function, antigen presentation and cytokine response have been associated with PI-IBS development. However, it is unknown whether specific pathogens have unique effects on long-term alterations in gut physiology or different pathogens converge to cause common alterations resulting in similar phenotype. The role of microbial virulence and pathogenicity factors in development of PI-IBS is also largely unknown. Additionally, alterations in host gut sensation, motility, secretion, and barrier function in PI-IBS need to be elucidated. Finally, both GI infections and antibiotics used to treat these infections can cause long-term alterations in host commensal microbiota. It is plausible that alteration in the commensal microbiome persists in a subset of patients predisposing them to develop PI-IBS.